CE HOU (ALBERT)

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EDUCATION

Hong Kong University of Science and Technology — Hong Kong SAR, China
Doctor of Philosophy (PhD) in Civil Engineering — Feb 2023 – Now

- Research interest: Geospatial artificial intelligence (GeoAI); AI for SDG; urban sensing; sustainable urban development
- Advisors: Prof. Fan Zhang and Prof. Sen Li

University College London — London, United Kingdom
Master of Science — Sep 2021 – Sep 2022

- Major in Spatial Data Science and Visualization
- Advisor: Prof. Chen Zhong

Wuhan University — Wuhan, China
Bachelor of Engineering — Sep 2017 – Jun 2021

- Major in Geodesy and Geomatics Engineering
- Advisors: Prof. Bao Zhang and Prof. Yibin Yao

VISITING AND RESEARCH EXPERIENCE

University of California, Berkeley — CA, United States
Visiting PhD student — Mar 2025 – Sep 2025

- Advisor: Prof. Lu Liang

Peking University — Beijing, China
Visiting PhD student — Feb 2024 – Now

- Advisors: Prof. Fan Zhang, Prof. Lei Dong, and Prof. Yu Liu

Tsinghua University — Beijing, China
Research Assistant — Jul 2022 – Dec 2022

- Advisor: Prof. Zhu Liu

The Bartlett School of Sustainable Construction, UCL — London, United Kingdom
Research Assistant — May 2022 – May 2023

- Advisors: Prof. Zhifu Mi and Dr. Haoran Zhang

PUBLICATIONS

Journal articles:

1. **Ce Hou**, Chen Zhong*, Huanfa Chen, Yuhao Kang, Shiqi Wang, Yikang Wang, Fan Zhang (2026). Why people cycle? Uncovering the mismatch between cycling visual environment and cycling behavior. *Transactions in Urban Data, Science, and Technology.* https://doi.org/10.1177/27541231261416070
2. Yecheng Zhang, Rong Zhao, Zhizhou Sha, Yong Li, Lei Wang, **Ce Hou**, Wen Ji, Hao Huang, Yunshan Wan, Jian Yu, Junhao Xia, Yuru Zhang, Chunlei Shi (2026). UrbanAlign: Post-hoc Semantic Calibration for VLM-Human Preference Alignment. *arXiv preprint arXiv:2602.19442*
3. Maoran Sun, **Ce Hou**, Qiaosi Li, Fan Zhang, Ronita Bardhan, Qunshan Zhao (2026). Deciphering Exterior: Building Energy Efficiency Prediction with Emerging Urban Big Data. *npj urban Sustainability.* https://doi.org/10.1038/s42949-026-00348-7
4. Qianbao Hou, **Ce Hou**, Fan Zhang, Qihao Weng (2025). Multi-source geo-localization in urban built environments for crowd-sourced images by contrastive learning. *ISPRS Journal of Photogrammetry and Remote Sensing*. https://doi.org/10.1016/j.isprsjprs.2025.09.024
5. Furong Jia, Lanxin Liu, **Ce Hou**, Fan Zhang*, Xinyan Liu, Yu Liu (2025). Towards Interpretable Geo-localization: a Concept-Aware Global Image-GPS Alignment Framework. *arXiv preprint arXiv:2509.01910*.
6. **Ce Hou**, Fan Zhang*, Yuhao Kang, Zhuangyuan Fan, Sen Li (2025). Transferring population group knowledge from multimodal large language model to small model: using urban safety

perception evaluation as case study, 07 May 2025, PREPRINT (Version 1) available at Research Square [https://doi.org/10.21203/rs.3.rs-6351052/v1]

7. **Ce Hou**, Fan Zhang*, Yong Li, Haifeng Li, Gengchen Mai, Yuhao Kang, Ling Yao, Wenhao Yu, Yao Yao, Song Gao, Min Chen, Yu Liu (2025). Urban Sensing in the Era of Large Language Models. *The innovation.* 6(1). DOI: https://doi.org/10.1016/j.xinn.2024.100749
8. **Ce Hou**, Yuhao Kang, Fan Zhang*, Song Gao, Yong Li, Fabio Duarte, Sen Li (2024). "Transferred bias" uncovers the balance between the development of physical and socio-economic environments of cities. *Annals of the American Association of Geographers*. https://doi.org/10.1080/24694452.2024.2412173
9. Lei Wang, **Ce Hou**, Yecheng Zhang, Jie He* (2024). Measuring solar radiation and spatio-temporal distribution in different street network direction through solar trajectories and street view images. *International Journal of Applied Earth Observation and Geoinformation*. https://doi.org/10.1016/j.jag.2024.104058
10. Piyu Ke, Xiaofan Gui, Wei Cao, Dezhi Wang, **Ce Hou**, Lixing Wang, Xuanren Song, Yun Li, Biqing Zhu, Jiang Bian, Stephen Sitch, Philippe Ciais, Pierre Friedlingstein, and Zhu Liu* (2023). Near-real-time monitoring of global ocean carbon sink. arXiv preprint arXiv:2312.01637, 2023.
11. Yikang Wang, Yuhao Kang, Haokun Liu, **Ce Hou**, Bing Zhou, Shan Ye, Yuyan Liu, Jinmeng Rao, Zhenghao Pei, Xiang Ye, Song Gao (2023). Choosing GIS graduate programs from afar: Chinese students' perspectives. *Transactions in GIS*. DOI: https://doi.org/10.1111/tgis.13037

Book chapter:

1. **Ce Hou**, Yong Li, Fan Zhang* (2024). Sensing Urban Physical Environment with GeoAI and Street-Level Imagery. *Handbook of Geospatial Approaches to Sustainable Cities*.

HONORS

- Overseas Research Award, HKUST 2025
- Best Poster Award, the 2nd Research Summit of Urban Science and Human Dynamics 2024
- Creativity Award, Research Summit of Urban Science and Human Dynamics 2023
- Bezos/Musk Prize for Entrepreneurial Potential in Knight Frank Dubai Housing Data Hackathon 2022
- Excellent undergraduate graduation design, Wuhan University 2021
- Distinct Graduate Student, Wuhan University 2021
- Distinct Student, Wuhan University 2021
- Li Qinghai Scholarship for Outstanding Students in Geomatics 2021
- First Class Scholarship, Wuhan University 2021
- Distinct Student, Wuhan University 2020
- Outstanding student leaders 2020
- Second Class Scholarship, Wuhan University 2020
- Social activist Student, Wuhan University 2020
- Merit Student, Wuhan University 2019

INVITED TALKS

Wuhan, China

Invited talk Mar 2025

Invited talk at the School of Public Administration, China University of Geosciences (Wuhan)

GRANTS

- **Co-Investigator**, Research on the Modernization Path of Street Ecological Governance from a Human-Computer Collaboration Perspective, funded by the *Youth Fund Project of Humanities and Social Sciences, Ministry of Education, ¥80,000* Jun 2025 – Jun 2028

TEACHING EXPERIENCE

Geospatial Science & Technology for Smart City (CIVL4100U) Hong Kong SAR, China

Teaching assistant 2023-24 Fall

Introduction to Smart City Economics (CIVL4640) Hong Kong SAR, China
Teaching assistant 2024-25 Spring

COMMUNITY SERVICES

The Chinese Professional in Geographic Information Systems Online
BOD student member Jan 2026 – Now
Fulfilling board responsibilities and participate in CPGIS's management and development.

GISphere Online
Co-chair Mar 2022 – Mar 2024
Participate in organizing online seminars (GISalon) and people interview section (GIStory), assisting in managing the official WeChat account and publishing tweets.

The Chinese Professional in Geographic Information Systems Online
Volunteer Dec 2021 – Now
Assist in managing the official WeChat account and publishing tweets.

Journal reviews:
Computer, Environment and Urban System; International journal of Geographical Information Science; Scientific Data; Humanity and Social Science Communications; Information Geography; Transactions in GIS; Scientific Reports; Journal of Urban Technology; Journal of Asian Architecture and Building Engineering; Computational Urban Science